Sushovan kc

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Education

- **Advanced College of Engineering and Management** — Tribhuvan University
  Bachelor of Engineering, Computer Engineering — *April 2022 – April 2026*
- **Ambition academy** — High School
  Major: Physical Science — *July 2019 – July 2021*

Skills Summary

- **Languages**: Python, C, C++, JavaScript, SQL, HTML/CSS
- **Frameworks & Tools**: Django, Django REST Framework, React (Vite), FastAPI, TensorFlow, PyTorch, OpenCV, Git/GitHub, Docker, RESTful API Design
- **Database & Cloud**: PostgreSQL, MySQL, SQL Server, SQLite, AWS (IaaS/PaaS)
- **Core Competencies**: Machine Learning, Deep Learning, Computer Vision, Full-Stack Web Development, Agile Collaboration, Technical Documentation

Experience

- **Nepal Telecom limited** — Jawalakhel, Lalitpur
  Full Stack Developer (Internship) — *May 2026 - Aug 2026*
  - **Full-Stack Development**: Engineered end-to-end web applications by integrating Django, Django REST Framework, and React (Vite) with PostgreSQL/SQLite backends, delivering secure authentication, database management, and seamless frontend-backend communication via RESTful APIs.
  - **Vehicle Management System (VMS)**: Spearheaded the design and development of a full-scale Vehicle Management System, building end-to-end booking workflows, vehicle and driver management modules, admin dashboards, file upload handling, and automated approval/notification pipelines to streamline fleet operations and cut manual coordination overhead.

Academic Projects

- **Segmentation of Brain Hemorrhage using U-Net and Masked Autoencoders** — April 2025 - April 2026
  Major Project
  - Architected supervised brain hemorrhage segmentation pipelines using U-Net and U-Net++, and designed a self-supervised Masked Autoencoder (MAE) framework integrated with a U-Net decoder for CT scan analysis.
  - Benchmarked supervised versus self-supervised approaches through comparative analysis, demonstrating that the MAE-based method matched strong segmentation performance while training on substantially fewer labeled samples.
- **Enhancing Ophthalmic Diagnostics Using U-Net for Cataract Detection** — September 2024 - March 2025
  Minor project
  - Built a U-Net-based deep learning model for automated cataract detection and segmentation from ophthalmic images, enabling faster and more consistent diagnostic screening.
  - Improved model robustness and detection accuracy for early-stage diagnosis by designing and applying targeted preprocessing and data augmentation pipelines.

PERSONAL PROJECTS

- **Play2Code: Gamified C++ Learning Platform** — Feb 2026
  Play2Code — C++, Godot, FastAPI, Python, JSON
  - Designed and built a gamified RPG in Godot with an exploration map, battle system, and embedded code editor to teach C++ fundamentals through interactive, scenario-based challenges.
  - Built a FastAPI backend with a secure Sandbox Runner to execute real-time code compilation, automated test evaluation, and JSON-based data persistence.
- **Personal Portfolio Website** — 2025
  HTML, CSS, JavaScript, GitHub Pages
  - Designed and deployed a fully responsive personal portfolio website using HTML, CSS, and JavaScript to showcase projects, technical skills, and achievements.
  - Deployed via GitHub Pages to provide a centralized, always-available platform for project demonstrations and professional profiling.

ACHIEVEMENT

- **Protobytes Hackathon 2.0** Kathmandu,Nepal
  5th position out of 50+ teams *Feb 2026*
  - Ranked 5th out of 50+ competing teams in a 60-hour hackathon organized by PACE at Advanced College of Engineering and Management, delivering a functional solution under intense time constraints.

Professional Training & Certifications

- **LaTeX Technical Documentation Workshop** May 2024 - June 2024
  Advanced College of Engineering and Management (ACEM)
  - Completed an intensive, hands-on workshop on LaTeX-based academic and technical documentation.
  - Produced professionally formatted research papers, technical reports, and presentations adhering to IEEE-compliant standards.
  - Gained hands-on proficiency in mathematical typesetting, figures, tables, cross-referencing, and BibTeX-based citation management.

- **AWS Cloud Computing Training** July 2025
  Advanced College of Engineering and Management (ACEM) — Prixa Academy
  - Completed a one-month, hands-on training program on Amazon Web Services (AWS) cloud computing.
  - Gained practical, applied knowledge of cloud architecture, deployment models (IaaS, PaaS), and core AWS services.
  - Designed and evaluated scalable, secure, and cost-efficient cloud-based solutions relevant to modern software and AI system deployment.